

Phase 8 Publishability Upgrade: External Validation, Benchmarks, and Gene/Pathway Sheaf Readiness

Sheaf-Theoretic Multi-Omics Brain Tumor Project

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Abstract

This document records the next concrete upgrade toward an IEEE BIBM-quality paper. Phase 8 adds a locked internal validation audit, an external CGGA validation adapter, a gene/pathway-level regulatory sheaf scaffold, benchmark requirements, and a manuscript-safe claim plan. No external CGGA performance is claimed here because raw CGGA files were not present in the workspace.

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1 Why Phase 8 Exists

Phases 1–7 established a biologically constrained sheaf residual framework for glioma multi-omics. The strongest current contribution is methodological: tumor states are analyzed by residual energies induced by sheaf restriction maps, rather than by ordinary feature fusion alone. For publication, the project needs three upgrades:

1. external validation, preferably TCGA-to-CGGA;
2. strict benchmark discipline against strong clinical, molecular, and graph-learning baselines;
3. higher biological resolution through gene/pathway-local residuals.

2 Core Method to Preserve

Let $x_{p,v} \in \mathcal{F}(v)$ be patient p 's state at biological node v . For an edge $e = (u, v)$, define

$$r_{p,e} = \rho_{u,e}x_{p,u} - \rho_{v,e}x_{p,v}. \quad (1)$$

The global sheaf regulatory inconsistency score is

$$\text{SRIS}(p) = \sum_{e \in E} \|r_{p,e}\|_2^2 = x_p^T L_{\mathcal{F}} x_p, \quad (2)$$

where $L_{\mathcal{F}} = B_{\mathcal{F}}^T B_{\mathcal{F}}$ is the sheaf Laplacian.

3 Phase 8 Locked Internal Audit

A locked 25% holdout split was used as an internal publication-readiness audit. This is not a substitute for external validation, but it helps detect overfitting and claim inflation before CGGA is added. The evaluated feature families were:

- strict clinical/molecular baseline;
- baseline plus Phase 1 SRIS and edge residuals;
- baseline plus Phase 4 counterfactual sheaf energies;
- baseline plus Phase 5 transport sheaf features;
- baseline plus Phase 6 consensus discovery index;
- baseline plus all sheaf feature families.

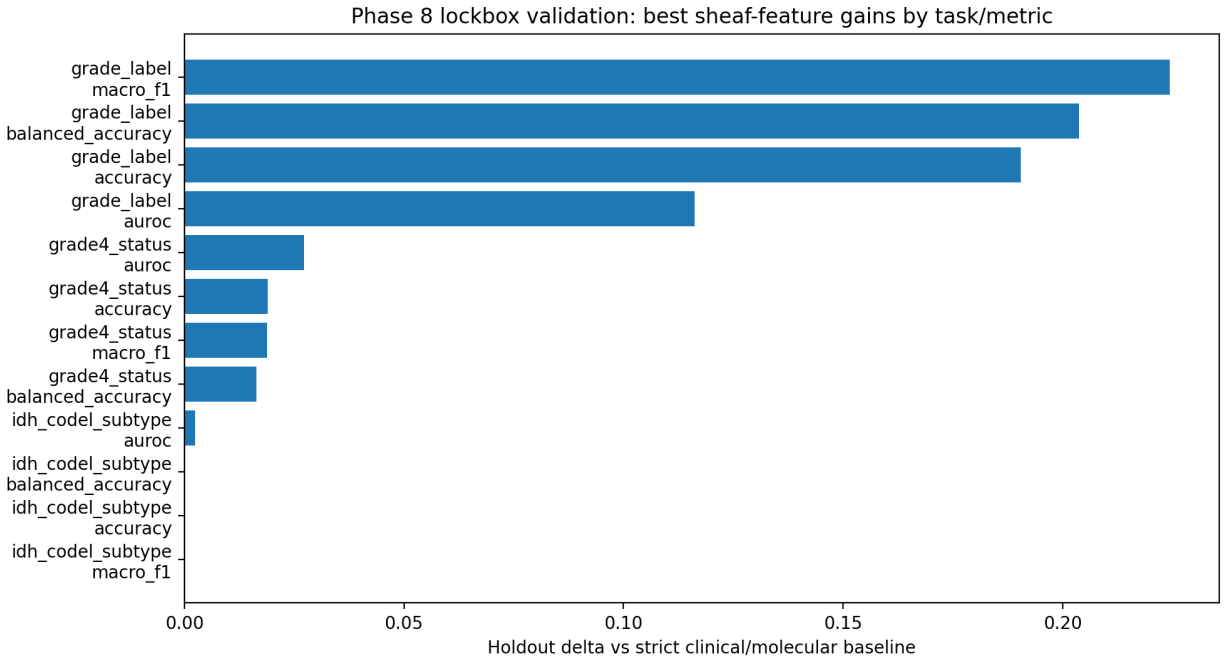


Figure 1: Best locked-holdout metric deltas versus the strict baseline. Positive bars indicate that a sheaf-derived feature family improved that metric on the held-out split.

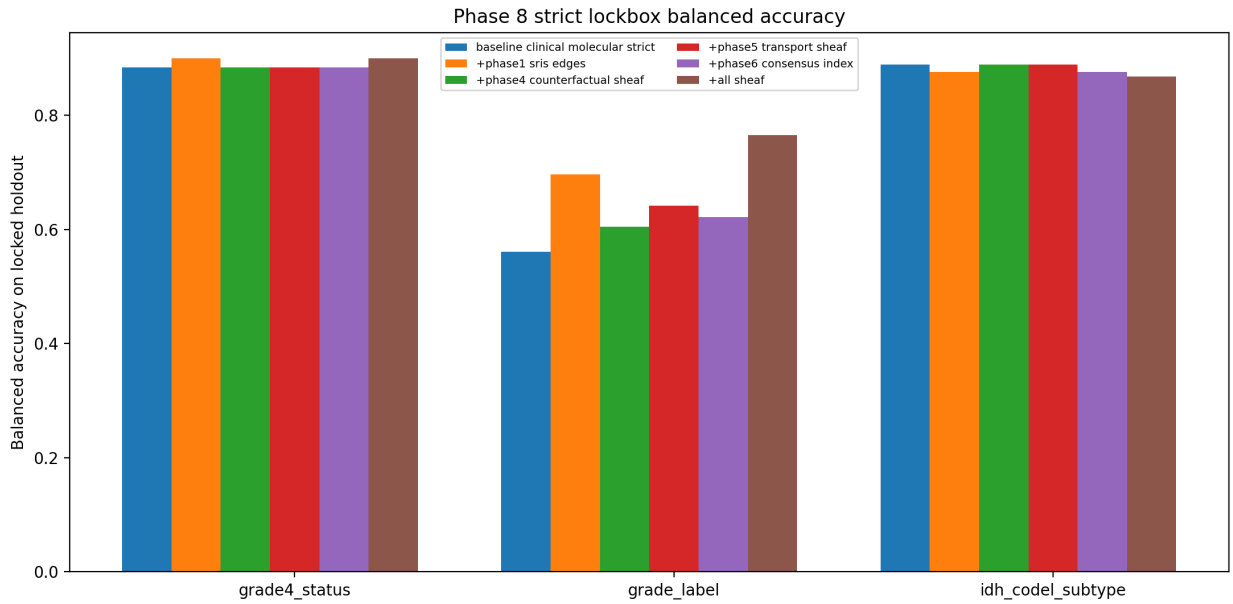


Figure 2: Balanced accuracy by task and feature family on the locked internal holdout.

4 External Validation Protocol

The decisive validation should train sheaf maps on TCGA and apply them to CGGA without refitting on CGGA labels. For external patient q , compute

$$\text{SRIS}_{\text{CGGA}}(q) = \sum_{e \in E} \|\rho_{u,e}^{\text{TCGA}} x_{q,u} - \rho_{v,e}^{\text{TCGA}} x_{q,v}\|^2. \quad (3)$$

Required external tests:

1. external grade/subtype AUROC and balanced accuracy;
2. external survival C-index or time-dependent AUROC;
3. subgroup calibration by IDH/codel status;
4. overlap of top residual signatures across TCGA and CGGA;
5. OT stability of residual signatures across cohorts.

5 Gene/Pathway-Level Regulatory Sheaf

The next major novelty upgrade is to move from summary nodes to gene/pathway nodes. For gene g and patient p , define

$$x_{p,g} = [\text{RNA}_{p,g}, \text{Methylation}_{p,g}, \text{CNV}_{p,g}, \text{Mutation}_{p,g}, \text{miRNA}_{p,g}]^T. \quad (4)$$

For regulatory edge $e = (u, v)$,

$$r_{p,e} = \rho_{u,e} x_{p,u} - \rho_{v,e} x_{p,v}. \quad (5)$$

For pathway P ,

$$\text{SRIS}_P(p) = \sum_{e \in E(P)} \|r_{p,e}\|^2. \quad (6)$$

This creates pathway-local biological discovery outputs rather than only patient-level risk scores.

6 Benchmark Requirements

A BIBM-quality paper should compare against:

- clinical logistic/Cox baselines;
- clinical + molecular baselines;
- elastic-net logistic and Cox models;
- random forest or gradient boosting;
- graph baselines such as GCN/GAT/graph transformer if gene-level data are used;
- identity, random, and shuffled-target sheaf controls.

7 Manuscript-Safe Claims

Safe current claim:

We introduce a biologically constrained sheaf residual geometry for quantifying genomic-regulatory-phenotype inconsistency in glioma multi-omics, with non-random subtype/grade geometry and strict internal validation safeguards.

Unsafe until CGGA is run:

The method achieves clinical survival state of the art.

8 What Must Happen Before Submission

1. Add CGGA clinical and matched omics files.
2. Run the external adapter and TCGA-to-CGGA sheaf scoring.
3. Add pathway/gene-level residuals if matched omics are available.
4. Re-run all ablations and benchmark models on identical splits.
5. Update claim ledger using external metrics only.
6. Write final IEEE two-column manuscript from the skeleton.